

Operations Research II: Algorithms

Course Overview

Ling-Chieh Kung

Department of Information Management
National Taiwan University

Road map

- ▶ **Course overview.**
- ▶ The row and column views for a linear system (a system of linear equations).
- ▶ Using Gaussian elimination to solve $Ax = b$.
- ▶ Using Gauss-Jordan elimination to solve A^{-1} .
- ▶ Linear dependence and independence.

Course overview

- ▶ We have learned how to formulate mathematical programs.
 - ▶ And how to use MS Excel to solve some instances.
 - ▶ And applications.
- ▶ However...
 - ▶ How does a solver solve an instance with thousands of variables and thousands of constraints?
 - ▶ Why does it take (much) more time to solve an integer program than to solve a linear program?
 - ▶ Why linearization is important? Why not just formulating problems into nonlinear programs?
 - ▶ What should we do if MS Excel is not enough?
 - ▶ What should we do if no solver is enough?
- ▶ We must study **exact algorithms**, **advanced solvers**, and **heuristic algorithms** to answer the above questions.

Exact algorithms and advanced solvers

- ▶ In this course, we study some classic **exact algorithms** that solve mathematical programs.
 - ▶ An exact algorithm finds an optimal solution for a given problem regardless of the time it takes.
 - ▶ Week 2: **the simplex method** for linear programming.
 - ▶ Week 3: **the branch-and-bound algorithm** for integer programming.
 - ▶ Week 4: **gradient descent** and **Newton's method** for nonlinear programming.
- ▶ In each week, we also study how to use Gurobi Optimizer, an advanced solver that is much more powerful than MS Excel in solving mathematical programs.

Heuristic algorithms

- ▶ In many cases, formulating a mathematical program and invoking a solver are not enough.
 - ▶ Because the practice is too complicated.
 - ▶ We may need to design our own algorithms for our own problems.
 - ▶ The algorithms in many cases are **heuristic algorithms**, which finds a near-optimal solution in a reasonable amount of time.
- ▶ In Week 5, we give one case study to illustrate this idea.

Prerequisite: linear algebra

- ▶ The subjects are really fascinating.
- ▶ However, some knowledge in **linear algebra** is needed.
- ▶ Later we will introduce some fundamental ideas that are required in this course.
- ▶ Thanks to professor Gilbert Strang in Massachusetts Institute of Technology and Professor Argon Chen in National Taiwan University.
 - ▶ For professor Strang's wonderful textbook *Linear Algebra with its Applications* and professor Chen's excellent course with the same name.

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Linear equations ($n = 2$)

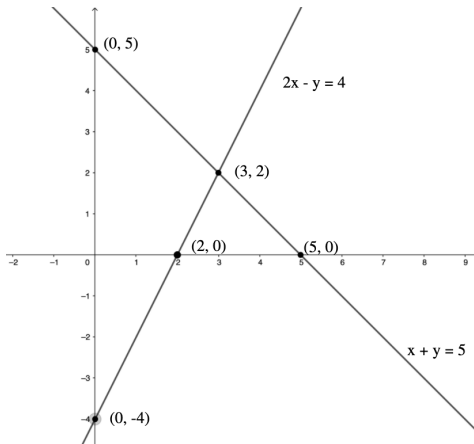
- ▶ Consider the following example:

$$x + y = 5$$

$$2x - y = 4.$$

- ▶ There are two perspectives to look at the system.
- ▶ By equation or **by row**: Each equation represents a **straight line** on the $x - y$ plane and the **intersection** is the only point on both lines and therefore the solution of the system.

Linear equations ($n = 2$)



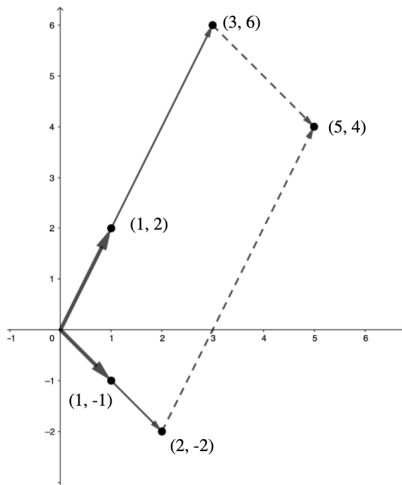
Linear equations ($n = 2$)

- **By column:** The two equations form one **vector equation**

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

We look for a **combination of the column vectors** on the left-hand side (LHS) which produces the vector on the right-hand side (RHS).

Linear equations ($n = 2$)



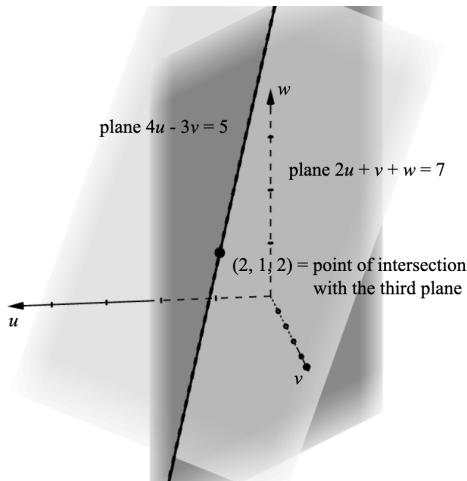
Linear equations ($n > 2$, by rows)

► Let's consider a 3-dimensional example:

$$\begin{array}{rcccccl} 2u & + & v & + & w & = & 7 \\ 4u & - & 3v & & & = & 5 \\ -2u & + & 4v & + & 2w & = & 4. \end{array}$$

Linear equations ($n > 2$, by rows)

- By rows: In **three** dimensions a **line** is formed by **two** equations; in n dimensions it is formed by $n - 1$ ones.



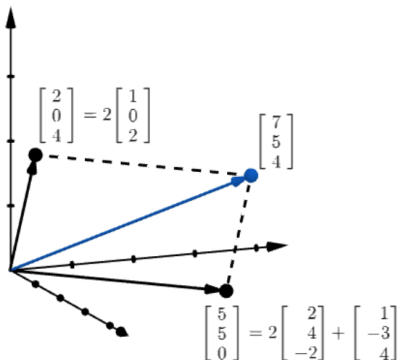
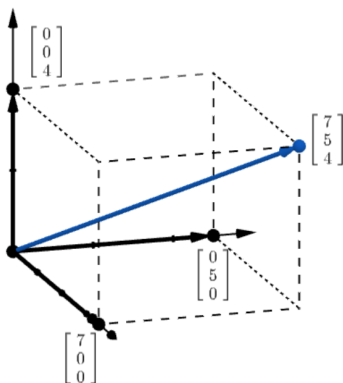
Linear equations ($n > 2$, by columns)

- By column: We still look for a way to combine the LHS columns to form the RHS column:

$$u \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} + w \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \\ 4 \end{bmatrix}.$$

Linear equations ($n > 2$, by columns)

► Visualization:



Linear equations ($n > 2$, by columns)

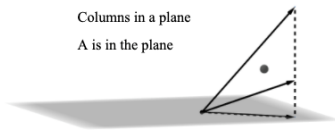
- ▶ In summary:
 - ▶ The **row view** considers a solution as an intersection of n (hyper)planes.
 - ▶ The **column view** considers a solution as a combination of the LHS column vectors to form the RHS column.
 - ▶ **Solution**: An “intersection of (hyper)planes” is a set of “coefficients in the combination of columns.”

Singular cases

- ▶ A linear system is called **singular** if there is no unique solution.
 - ▶ The row view: the n (hyper)planes do not intersect at exactly one point.



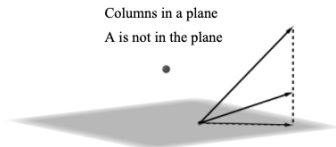
- ▶ The column view: the n vectors do not span a complete n -dimensional space.



Columns in a plane

A is in the plane

Infinity of solutions



Columns in a plane

A is not in the plane

No solution

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Gaussian elimination for solving $Ax = b$

- ▶ Let's solve this linear system

$$\begin{array}{rrcrcl} 2u & + & v & + & w & = & 7 \\ 4u & - & 3v & & & = & 5 \\ -2u & + & 4v & + & 2w & = & 4. \end{array}$$

- ▶ Let's illustrate **Gaussian elimination**, which is the most widely adopted way for solving linear systems.

Gaussian elimination for solving $Ax = b$

- ▶ Stage 1: **forward elimination**.
- ▶ We start by eliminating u from the last two equations by subtracting multiples of the first equation from the others. We obtain

$$\begin{array}{rrcrcl} 2u & + & v & + & w & = & 7 \\ & & -5v & - & 2w & = & -9 \\ & & 5v & + & 3w & = & 11. \end{array}$$

- ▶ 2: the **first pivot**.
- ▶ -5 : the **second pivot**.

Gaussian elimination for solving $Ax = b$

- ▶ We continue to eliminate v from the third equation.

$$\begin{array}{rcrcrcrcrcl} 2u & + & & v & + & w & = & 7 \\ & & & -5v & - & 2w & = & -9 \\ & & & & & w & = & 2. \end{array}$$

- ▶ **Pivots:** 2, -5 , 1.
- ▶ We then do stage 2, **back substitution** to first solve w , then v , and finally u .

Gaussian elimination for solving $Ax = b$

► We may do this by hands:

$$\begin{aligned} & \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 4 & -3 & 0 & 5 \\ -2 & 4 & 2 & 4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 0 & -5 & -2 & -9 \\ 0 & 5 & 3 & 11 \end{array} \right] \\ & \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 0 & -5 & -2 & -9 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 0 & -5 & 0 & -5 \\ 0 & 0 & 1 & 2 \end{array} \right] \\ & \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]. \end{aligned}$$

Singular cases

- ▶ When we have n equations, ideally we should obtain n pivots. If so, we say the system is **nonsingular**, and we will obtain exactly one solution.
- ▶ What if a zero appears in a pivot position? This may or may not be a problem.
- ▶ A nonsingular example (cured by row exchange).

$$\begin{array}{rclcl} 2u & + & v & + & w & = & & 2u & + & v & + & w & = \\ 4u & + & 2v & - & w & = & \rightarrow & & & - & 3w & = \\ -2u & + & 4v & + & 2w & = & & & & 5v & + & 3w & = \end{array}$$

$$\rightarrow \begin{array}{rclcl} 2u & + & v & + & w & = \\ & & 5v & + & 3w & = \\ & & & - & 3w & = \end{array}$$

Singular cases

- ▶ A **singular** example:

$$\begin{array}{rclcl}
 2u & + & v & + & w & = & & 2u & + & v & + & w & = \\
 4u & + & 2v & - & w & = & \rightarrow & & & & - & 3w & = \\
 -2u & - & v & + & 2w & = & & & & & 3w & =
 \end{array}$$

- ▶ What will happen if a system is singular?
 - ▶ If $-3w = -6$ and $3w = 7$, the system is **inconsistent**, and there is no solution.
 - ▶ If $-3w = -6$ and $3w = 6$, the system is **consistent**, and there are infinitely many solutions. For all of them, their $2u + v$ are identical.

Complexity of Gaussian elimination

- ▶ How many elementary arithmetical operations does Gaussian elimination require to solve a system with n equations and n variables?
- ▶ An “operation” may be a division, a multiplication-subtraction, etc.

Complexity of Gaussian elimination

- ▶ To complete the elimination for the first column:
 - ▶ To produce a zero in the first column: one division and n multiplication-subtraction (including the RHS) are needed. In total $n + 1$ operations are needed.
 - ▶ There are $n - 1$ rows to be processed: $(n + 1)(n - 1) = n^2 - 1$ operations are needed.
- ▶ For the second column: $(n - 1)^2 - 1$.
- ▶ For the third column: $(n - 2)^2 - 1$.
- ▶ There are n columns:

$$(n^2 + \dots + 1^2) - (1 + \dots + 1) = \frac{n(n+1)(2n+1)}{6} - n = \frac{n^3}{3} + \frac{n^2}{2} - \frac{5n}{6}.$$

- ▶ When n is large, it is roughly $\frac{n^3}{3}$.

Complexity of Gaussian elimination

- ▶ Number of operations required for back substitution is roughly

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \approx \frac{n^2}{2}.$$

- ▶ Technically, the **complexity** of Gaussian elimination is $O(n^3)$.
- ▶ The computation time needed to complete Gaussian elimination is proportional to n^3 .

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The inverse

- ▶ A matrix A is **invertible** if there exists a matrix B such that $BA = I$ and $AB = I$.
- ▶ Such a matrix B is called the **inverse** of A and denoted by A^{-1} , i.e.,

$$A^{-1}A = I \text{ and } AA^{-1} = I.$$

- ▶ A^{-1} is **unique** if it exists. To see this, note that if $BA = I$ and $AC = I$, we have

$$B = BI = B(AC) = (BA)C = IC = C.$$

Gauss-Jordan elimination for finding A^{-1}

- ▶ To find A^{-1} from A , we use **Gauss-Jordan elimination**, which simply does

$$\left[A \mid I \right] \rightarrow \left[I \mid A^{-1} \right].$$

- ▶ Before we explain why, let's see an example.

Gauss-Jordan elimination for finding A^{-1}

► An example:

$$\begin{aligned} \left[A \mid e_1 \quad e_2 \quad e_3 \right] &= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 4 & 3 & 5 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 5 & -4 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 6 & 1 & -5 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array} \right] \end{aligned}$$

Gauss-Jordan elimination for finding A^{-1}

- Continue from the previous page:

$$\begin{aligned} &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & 1 & -5 \\ 0 & -1 & 0 & 6 & 1 & -5 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & 1 & -5 \\ 0 & 1 & 0 & -6 & -1 & 5 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array} \right]. \end{aligned}$$

- We may verify that

$$\begin{bmatrix} 1 & 1 & 0 \\ 4 & 3 & 5 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 7 & 1 & -5 \\ -6 & -1 & 5 \\ -2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 1 & -5 \\ -6 & -1 & 5 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 4 & 3 & 5 \\ 2 & 2 & 1 \end{bmatrix},$$

and both products result in I .

Gauss-Jordan elimination for finding A^{-1}

- Why does this method work?

$$AA^{-1} = I \iff \begin{array}{rcl} Ax_1 & = & e_1 \\ Ax_2 & = & e_2 \\ Ax_3 & = & e_3 \end{array}, A^{-1} = \begin{bmatrix} \vdots & \vdots & \vdots \\ x_1 & x_2 & x_3 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$\begin{aligned} &\iff [A \mid e_1], [A \mid e_2], [A \mid e_3] \\ &\quad \rightarrow [I \mid x_1], [I \mid x_2], [I \mid x_3] \end{aligned}$$

$$\iff [A \mid I] \rightarrow [I \mid A^{-1}].$$

Some remarks

- ▶ The complexity of Gauss-Jordan elimination is also $O(n^3)$.
- ▶ A square matrix is **nonsingular** if and only if it is **invertible**.

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Linear dependence and independence

- ▶ A set of m n -dimensional vectors $x_1, x_2, \dots$, and x_m are **linearly dependent** if there exists a non-zero vector $w \in \mathbb{R}^m$ such that

$$w_1x_1 + w_2x_2 + \cdots w_mx_m = 0.$$

- ▶ Say it in another way: we may linearly combine $n - 1$ of the n vectors to generate the last one.
- ▶ A set of vectors are **linearly independent** if they are not linearly dependent.

- ▶ E.g., $\begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ are linearly independent.

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Linear dependence and independence

► Example 1:

$$\left[\begin{array}{cc|c} 2 & 1 & 1 \\ 4 & -3 & 0 \\ -2 & 4 & 2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 5 & 3 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & 1 \end{array} \right].$$

► Example 2:

$$\left[\begin{array}{cc|c} 2 & 1 & 1 \\ 4 & -3 & 0 \\ -2 & 4 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 5 & 2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & 0 \end{array} \right].$$

Linear dependence and independence

► Wait... how about this?

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -3 & 0 \\ -2 & 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

► And this?

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -3 & 0 \\ -2 & 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 5 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Linear dependence and independence

- ▶ In summary, just put all vectors into the columns of a matrix. Then use Gaussian elimination to find the number of **pivots** we have.
 - ▶ The number of pivots is the number of linearly independent vectors.
 - ▶ If the number of pivot **equals m** , the number of vectors, these vectors are all **linearly independent**.
 - ▶ If the number of pivot is **less than m** , they are **linearly dependent**.
- ▶ This implies that m n -dimensional vectors must be linearly dependent if $m > n$.
 - ▶ When $n > m$, however, these vectors are not necessarily linear independent.